Practical 1:-Configuring LAN setup

a. Planning. and Setting IP networks

b. Configuring subnet

c. Using linux-terminal or Windows-cmd, execute following networking commands and note the output: ping, traceroute, netstat, arp, ipconfig, Getmac, hostname, NSLookUp, pathping, SystemInfo.

Aim:-To Configure a LAN and test Connectivity.

Apparatus:- Cisco Packet Tracer, PC0,PC1, Switch 2960, cables.

Procedure:- 1.Connect devices

2.Assign IP

3.Test ping

4.Run Supported commands.

A) Planning and Setting IP networks

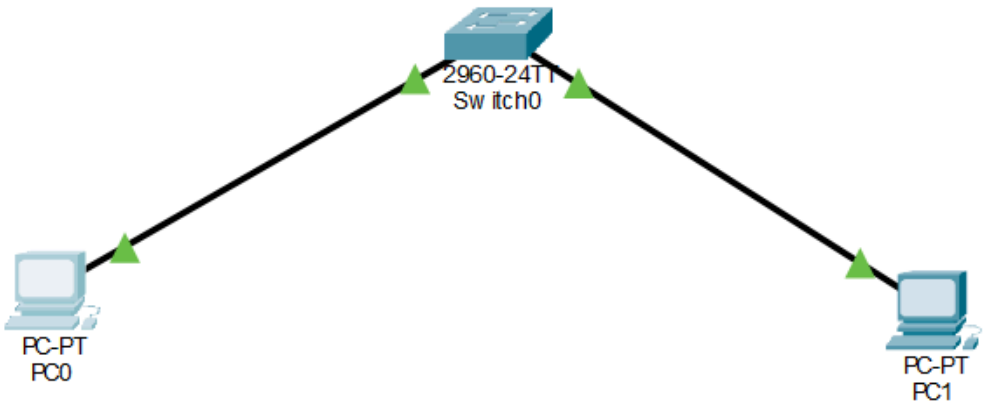
PC0

PC1

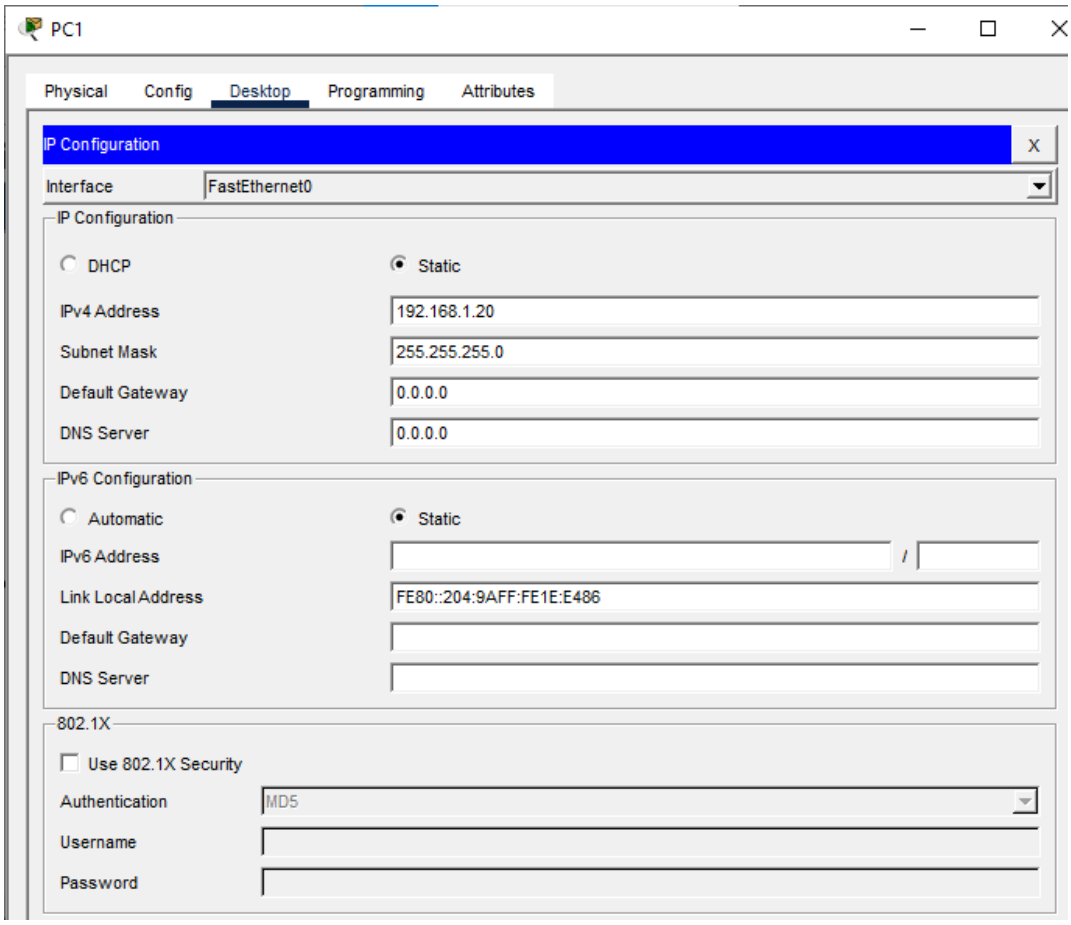
Switch 2960

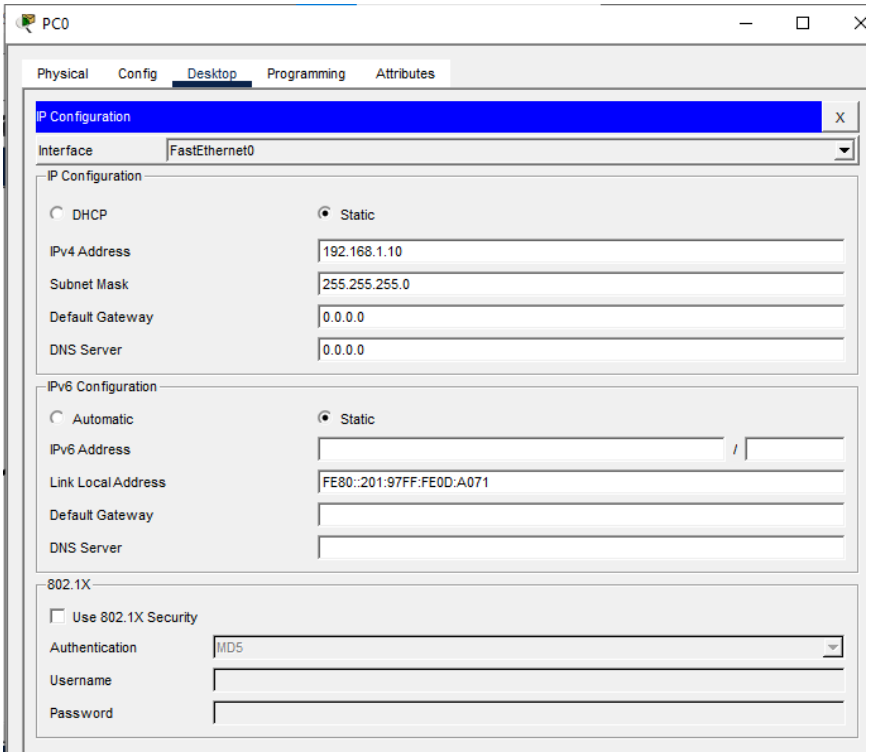
Copper straight-through cables

Output:-



B) Configuring subnet





Hu u

C) Using linux-terminal or Windows-cmd, execute following networking commands and note the output: ping, traceroute, netstat, arp, ipconfig, Getmac, hostname, NSLookUp, pathping, SystemInfo.

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:97FF:FE0D:A071
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.20          0004.9a1e.e486        dynamic

C:\>tracert 192.168.1.20

Tracing route to 192.168.1.20 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.20

Trace complete.

C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
C:\>hostname
Invalid Command.

C:\>getmac
Invalid Command.

C:\>systeminfo
Invalid Command.

C:\>nslookup

Server: [255.255.255.255]
Address:   255.255.255.255

>pathping 192.168.1.20
Server: [192.168.1.20]
Address:  192.168.1.20
DNS request timed out.
            timeout was 15000 milli seconds.
DNS request timed out.
            timeout was 15000 milli seconds.
DNS request timed out.
            timeout was 15000 milli seconds.
*** Request to 192.168.1.20 timed-out
```

Practical 2:- IPv4 Addressing and Subnetting.

IPv4 Addressing and Subnetting

1. **Given an IP address and a network mask**, determine the following information:

   a. **Network address**
   b. **Network broadcast address**
   c. **Total number of host bits**
   d. **Total number of usable hosts**

2. **Given an IP address and a subnet mask**, determine the following information for the subnet:

   a. **Subnet address**
   b. **Subnet broadcast address**
   c. **Range of usable host addresses within the subnet**
   d. **Maximum number of subnets created by this subnet mask**
   e. **Number of hosts per subnet**
   f. **Number of subnet bits used**
   g. **The specific subnet number for the given IP address**

AIM:-To study IPv4 addressing and subnetting.
To determine the **network address, broadcast address, number of hosts, subnet mask details, subnet range**, and to verify connectivity using Cisco Packet Tracer.

APPARATUS / SOFTWARE REQUIRED

1. **Cisco Packet Tracer** (any version 7.x or above)

2. PC, Laptop or Lab Computer

3. Calculator for binary conversion (optional)

4. Networking commands (ipconfig, ping, tracert, arp, etc.)

5. Optional physical devices (only for explanation):

    - 2 PCs

    - 1 Switch

    - 1 Router

THEORY

1. Network Address

The first address in a network; all host bits = 0.

2. Broadcast Address

The last address in a network; all host bits = 1.

3. Number of Hosts

Formula: **2^(Host bits) – 2**

4. Number of Subnets

Formula: **2^(Borrowed bits)**

5. Host Range

From first usable host → last usable host.

PROCEDURE (MANUAL CALCULATION)

Assume the given IP address example: **192.168.10.25/26**

1. Convert prefix `/26` to subnet mask →
   /26 = 255.255.255.192

2. Determine **block size** →
   256 – 192 = 64

3. Determine **subnet range**:
   0–63, 64–127, 128–191, 192–255

4. Find the block in which *25* belongs →
   falls in **0–63**

5. Network Address = **192.168.10.0**

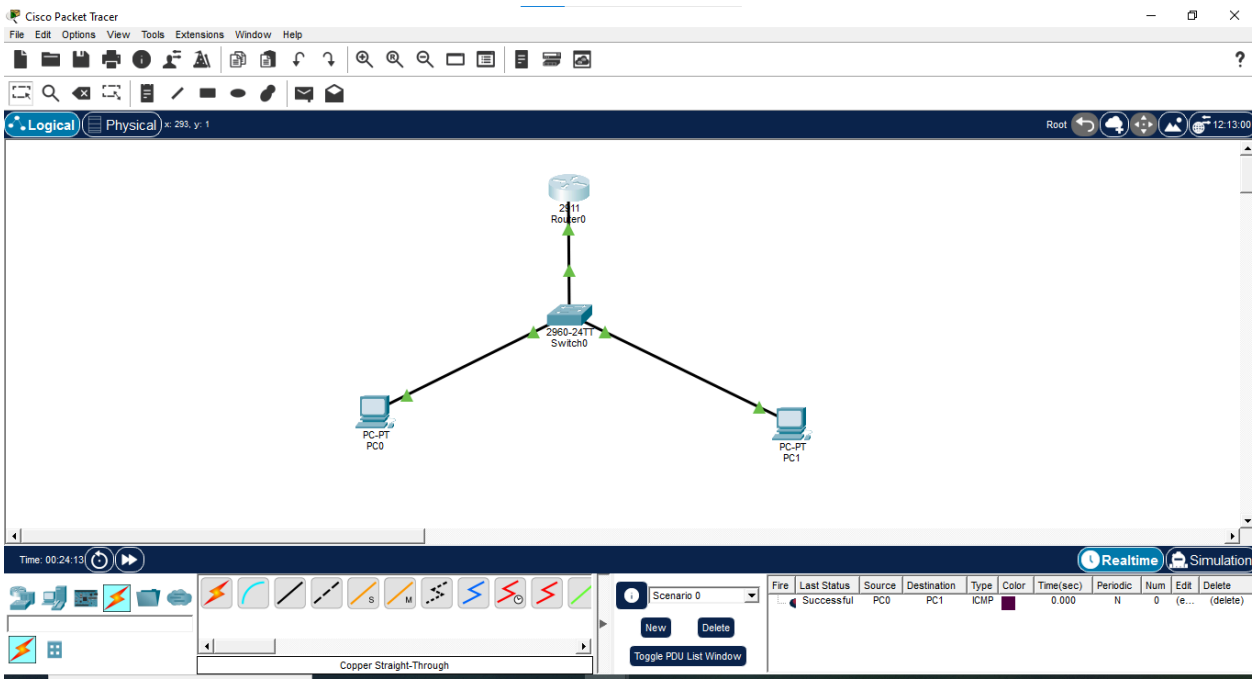
6. Broadcast Address = **192.168.10.63**

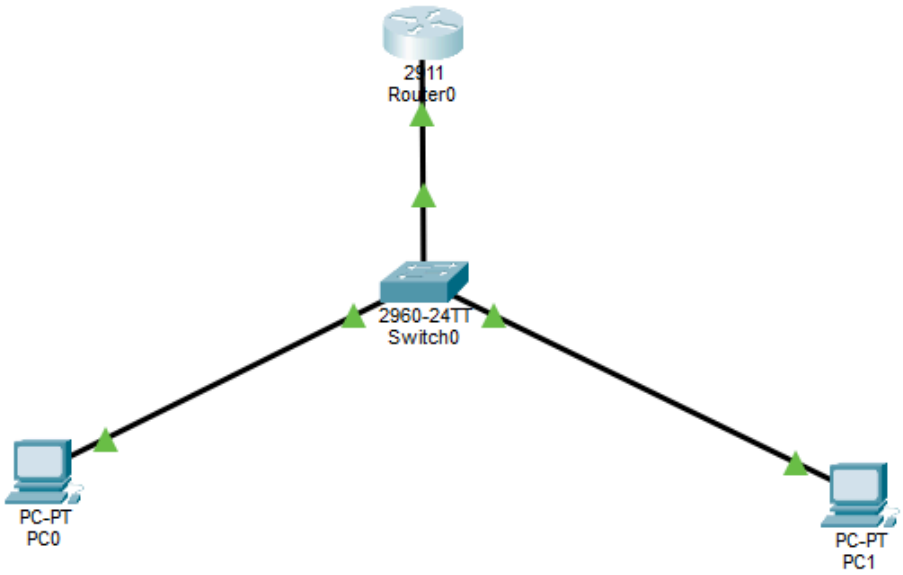
7. First Host = **192.168.10.1**

8. Last Host = **192.168.10.62**

9. Number of Hosts = 2^(32–26) – 2 = 62

10. Number of Subnets (Class C) = 2^(26–24) = 4




2911
Router0
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1

Router0

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip address 192.168.10.1 255.255.255.0
                   ^
% Invalid input detected at '^' marker.

Router(config)# ip address 192.168.10.1 255.255.255.0
                    ^
% Invalid input detected at '^' marker.

Router(config)# ip adress 192.168.10.1 255.255.255.0
                    ^
% Invalid input detected at '^' marker.

Router(config)# ip address 192.168.10.1 255.255.255.0
                    ^
% Invalid input detected at '^' marker.

Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy | Paste

Top

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=13ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::230:F2FF:FE8C:A849
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.0
```

Top

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::230:F2FF:FE8C:A849
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>tracert 192.168.10.20

Tracing route to 192.168.10.20 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.20

Trace complete.

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.1          0030.f252.b501        dynamic
```

Top

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>tracert 192.168.10.20

Tracing route to 192.168.10.20 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.20

Trace complete.

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.1          0030.f252.b501        dynamic
  192.168.10.20         0030.a3d1.de01        dynamic

C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
C:\>nslookup

Server: [255.255.255.255]
Address:   255.255.255.255

>getmac
Server: [255.255.255.255]
Address:   255.255.255.255
DNS request timed out.
           timeout was 15000 milli seconds.
DNS request timed out.
           timeout was 15000 milli seconds.
DNS request timed out.
           timeout was 15000 milli seconds.
*** Request to 0.0.0.0 timed-out


>systeminfo
Server: [255.255.255.255]
Address:   255.255.255.255
```

Top

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
  Internet Address      Physical Address      Type
  192.168.10.1          0030.f252.b501        dynamic
  192.168.10.20         0030.a3d1.de01        dynamic

C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
C:\>nslookup

Server: [255.255.255.255]
Address:   255.255.255.255

>getmac
Server: [255.255.255.255]
Address:   255.255.255.255
DNS request timed out.
           timeout was 15000 milli seconds.
DNS request timed out.
           timeout was 15000 milli seconds.
DNS request timed out.
           timeout was 15000 milli seconds.
*** Request to 0.0.0.0 timed-out


>systeminfo
Server: [255.255.255.255]
Address:   255.255.255.255
DNS request timed out.
           timeout was 15000 milli seconds.
DNS request timed out.
           timeout was 15000 milli seconds.
DNS request timed out.
           timeout was 15000 milli seconds.
*** Request to 0.0.0.0 timed-out


>
```

Top

🔵 EXPERIMENT 3: Configure Static IP Routing using Cisco Packet Tracer

Aim

To configure and verify static IP routing between multiple networks using Cisco Packet Tracer.

Objectives

- To understand manual routing configuration.
- To create static routes for unreachable networks.
- To verify end-to-end communication.

Tools Required

- Cisco Packet Tracer
- 2 or 3 Routers, PCs, Switches, Cables

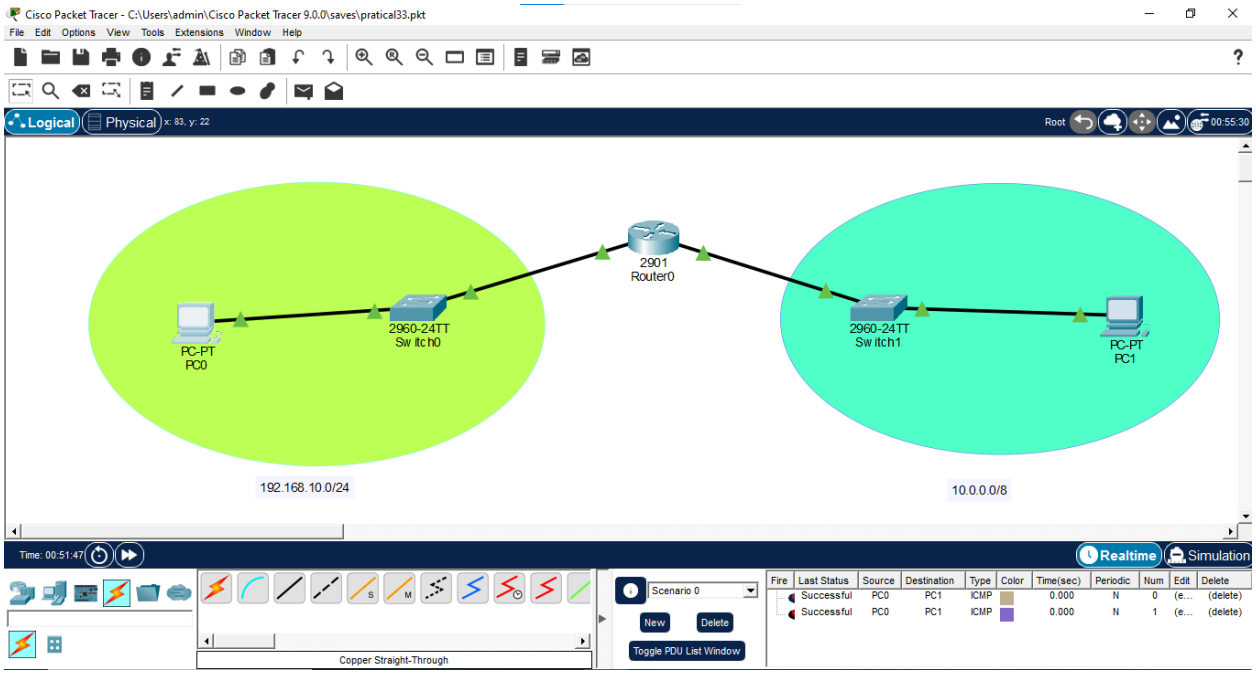
Network Topology (Example)

R1 —— R2 —— R3

Each router connected to its own LAN.

Procedure

1. Create a topology using routers, switches, and PCs.
2. Assign IP addresses to all PCs and router interfaces.
3. Enable interfaces with `no shutdown`.
4. Configure static routes:


Cisco Packet Tracer - C:\Users\admin\Cisco Packet Tracer 9.0.0\saves\pratical33.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
00:55:30
2901
Router0
PC-PT
PC0
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC1
192.168.10.0/24
10.0.0.0/8
Time: 00:51:47
Realtime
Simulation
Copper Straight-Through
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 PC1 ICMP 0.000 N 0 (e... (delete)
Successful PC0 PC1 ICMP 0.000 N 1 (e... (delete)

PC0

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

Automatic | Static

IPv6 Address: /

Link Local Address: FE80::2D0:97FF:FED8:63ED

Default Gateway:

DNS Server:

802.1X

Use 802.1X Security

Authentication: MD5

Username:

Password:

Top

PC1

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- DHCP
- Static

IPv4 Address	10.0.0.10
Subnet Mask	255.0.0.0
Default Gateway	10.0.0.1
DNS Server	0.0.0.0

IPv6 Configuration

- Automatic
- Static

IPv6 Address	/
Link Local Address	FE80::20C:85FF:FEC2:B4DE
Default Gateway	
DNS Server	

802.1X

- Use 802.1X Security

Authentication	MD5
Username	
Password	

Top

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigaEthernet0/0
                                ^
% Invalid input detected at '^' marker.

Router(config)#interface gigabitethernet 0/0
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up

Router(config-if)#
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.10 255.255.255.0
Router(config-if)#%IP-4-DUPADDR: Duplicate address 192.168.10.10 on
GigabitEthernet0/0, sourced by 00D0.97D8.63ED

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#%IP-4-DUPADDR: Duplicate address 192.168.10.10 on
GigabitEthernet0/0, sourced by 00D0.97D8.63ED

Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#
Router(config-if)#
```